

ECE F331: Communication Systems
Notes + Problems

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Chapter 1

Module 1

1.1 Classroom Problems from Slides

Question 1

A digit bit '1' can be represented by a rectangular pulse of width 1 second and the digit bit '0' can be represented by with no signal of pulse width 1 second,. Compute everage energy per bit E_b .

Solution:

Given,

Pulse width $T = 1$ second

Amplitude of pulse $A = 1$ V

Energy per bit $E_b = \int_0^T |s(t)|^2 dt$

where $s(t)$ is the signal representing bit '1'

Therefore, $E_b = \int_0^1 1^2 dt = 1$ J

Question 2

Energy of signal $m(t) = 100\text{sinc}(200\pi t)$?

Solution:

Given,

$m(t) = 100\text{sinc}(200\pi t)$

We know that, $E = \int_{-\infty}^{\infty} |m(t)|^2 dt$

Therefore, $E = \int_{-\infty}^{\infty} 10000 \sin^2(200\pi t) dt$

$= 10000 \int_{-\infty}^{\infty} \sin^2(200\pi t) dt$

$= 10000 \times \infty$ (since $\int_{-\infty}^{\infty} \sin^2(at) dt$ diverges)

Hence, the energy of the signal is infinite.

Question 3

Find the bandwidth of the signal $m(t) = 100\text{sinc}(200\pi t)$

Solution:

We know that the fourier transform of sinc is Π rect function.

Therefore, the fourier transform of $m(t)$ is given by,

$M(f) = \Pi\left(\frac{f}{200}\right)$ As such, the bandwidth of the signal is 100 Hz.

Question 4

Find the spectral density of the signal $m(t) = A \cos(2\pi f_c t + \theta)$

Solution:

Spectral density is given by $\psi(f) = |M(f)|^2$

We know that the fourier transform of $m(t)$ is given by,

$$M(f) = \frac{A}{2}[\delta(f - f_c) + \delta(f + f_c)]$$

Therefore, the spectral density is given by,

$$\begin{aligned}\psi(f) &= \left| \frac{A}{2}[\delta(f - f_c) + \delta(f + f_c)] \right|^2 \\ &= \frac{A^2}{4}[\delta(f - f_c) + \delta(f + f_c)]^2\end{aligned}$$

Question 5: Textbook Question

Show that the autocorrelation function of $g(t) = C \cos(2\pi f_0 t + \theta_0)$ is given by $\mathbb{R}_g(\tau) = \frac{C^2}{2} \cos(2\pi f_0 \tau)$, and the corresponding PSD is given by $\psi_g(f) = \frac{C^2}{4}[\delta(f - f_0) + \delta(f + f_0)]$. Hence, show that for a signal $y(t)$ given by:

$$y(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(2\pi f_n t + \theta_n)$$

the autocorrelation function is given by:

$$\begin{aligned}\mathbb{R}_y(\tau) &= C_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} C_n^2 \cos(2\pi f_0 \tau) \\ \psi_y(f) &= C_0^2 \delta(f) + \frac{1}{4} \sum_{n=1}^{\infty} C_n^2 [\delta(f - f_n) + \delta(f + f_n)]\end{aligned}$$

Solution:

We know that the autocorrelation function is given by,

$$\mathbb{R}_g(\tau) = E[g(t)g(t + \tau)]$$

$$= E[C \cos(2\pi f_0 t + \theta_0) C \cos(2\pi f_0 (t + \tau) + \theta_0)]$$

$$= C^2 E[\cos(2\pi f_0 t + \theta_0) \cos(2\pi f_0 t + 2\pi f_0 \tau + \theta_0)]$$

Using the trigonometric identity, $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, we get,

$$\mathbb{R}_g(\tau) = C^2 E \left[\frac{1}{2} [\cos(-2\pi f_0 \tau) + \cos(4\pi f_0 t + 2\theta_0 + 2\pi f_0 \tau)] \right]$$

$$= \frac{C^2}{2} [\cos(2\pi f_0 \tau) + E[\cos(4\pi f_0 t + 2\theta_0 + 2\pi f_0 \tau)]]$$

Since θ_0 is uniformly distributed over $[0, 2\pi]$, $E[\cos(4\pi f_0 t + 2\theta_0 + 2\pi f_0 \tau)] = 0$. Therefore,

$$\mathbb{R}_g(\tau) = \frac{C^2}{2} \cos(2\pi f_0 \tau)$$

Now, the PSD is given by the fourier transform of the autocorrelation function. Therefore,

$$\psi_g(f) = \int_{-\infty}^{\infty} \mathbb{R}_g(\tau) e^{-j2\pi f \tau} d\tau$$

$$= \int_{-\infty}^{\infty} \frac{C^2}{2} \cos(2\pi f_0 \tau) e^{-j2\pi f \tau} d\tau$$

Using the identity, $\cos x = \frac{e^{jx} + e^{-jx}}{2}$, we get,

$$\psi_g(f) = \frac{C^2}{4} \int_{-\infty}^{\infty} [e^{j2\pi f_0 \tau} + e^{-j2\pi f_0 \tau}] e^{-j2\pi f \tau} d\tau$$

$$= \frac{C^2}{4} \left[\int_{-\infty}^{\infty} e^{-j2\pi(f-f_0)\tau} d\tau + \int_{-\infty}^{\infty} e^{-j2\pi(f+f_0)\tau} d\tau \right]$$

Using the property of delta function, $\int_{-\infty}^{\infty} e^{-j2\pi a t} dt = \delta(a)$, we get,

$$\psi_g(f) = \frac{C^2}{4} [\delta(f - f_0) + \delta(f + f_0)]$$

Now, for the signal $y(t)$, the autocorrelation function is given by,

$$\mathbb{R}_y(\tau) = E[y(t)y(t + \tau)]$$

$$= E \left[\left(C_0 + \sum_{n=1}^{\infty} C_n \cos(2\pi f_n t + \theta_n) \right) \left(C_0 + \sum_{n=1}^{\infty} C_n \cos(2\pi f_n (t + \tau) + \theta_n) \right) \right]$$

$$= E \left[C_0^2 + C_0 \sum_{n=1}^{\infty} C_n \cos(2\pi f_n (t + \tau) + \theta_n) + C_0 \sum_{n=1}^{\infty} C_n \cos(2\pi f_n t + \theta_n) + \sum_{n=1}^{\infty} C_n^2 \cos(2\pi f_n t + \theta_n) \cos(2\pi f_n (t + \tau) + \theta_n) \right]$$

Since θ_n are uniformly distributed over $[0, 2\pi]$, $E[\cos(2\pi f_n t + \theta_n)] = 0$. Therefore,
 $\mathbf{R}_y(\tau) = C_0^2 + \sum_{n=1}^{\infty} C_n^2 E[\cos(2\pi f_n t + \theta_n) \cos(2\pi f_n(t + \tau) + \theta_n)]$
 Using the result from the first part, we get,
 $\mathbf{R}_y(\tau) = C_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} C_n^2 \cos(2\pi f_n \tau)$